Open problems:
Non-linear representations in quantum mechanics

Open problems raised by the technical brief in this directory, and by its Lean formalization (see `formalization-log.tex`). Items marked [BLOCKS VERIFICATION] stand in the way of checking a main claim of the brief as the brief states it: a *formalization gap* can be closed in Lean alone, while a *brief issue* needs the brief to be corrected or made precise first.

From the brief

1. **Transport of the rest of quantum mechanics.** The brief claims that "every other relationship in quantum mechanics can be expressed in this non-linear space through m", including superpositions, expectations of observables and time evolution. Only rays and the Born rule are checked. (Brief, Sec. 3, Remark.)

2. **No benefit from non-linear representations.** The transported relationships "would become considerably more complicated for likely no benefit". This is a speculation; no criterion for "benefit" is given. (Brief, Sec. 3, Remark.)

3. **Madelung representation.** "The Madelung equations ... provide such a representation", one "not even tied to vector spaces". The equivalence of the Madelung equations with quantum theory, and in what sense they represent QST, are not shown. (Brief, Sec. 4.)

4. **Any dimension.** The proof needs a non-linear, Born-rule-preserving, ray-preserving map on the $\mathcal{H}$ of the system, but constructs m only for a two-state system. The same formula $m(\psi) = e^{\imath\theta\, p(z^+|\psi)}\psi$ for a unit vector $|z^+\rangle$ should work whenever $\dim \mathcal{H} \geq 2$, including infinite dimension; this is not formalized ($\mathcal{H} = \mathbb{C}^2$ only). (Brief, Sec. 3, eq. (2); Log, "Setting".)

From the formalization

1. [BLOCKS VERIFICATION: BRIEF ISSUE] **Theorem 1 as an independence statement.** "REP-VSLIN is independent of QST" is a statement about logical independence, but ¬REP-VSLIN is not made precise, and its truth depends on the reading.

   - *Literal independence over models.* If a system is only its state space and probability, REP-VSLIN follows from QST (take the H that QST provides), so ¬REP-VSLIN is inconsistent with QST. If a system also carries vector-level data that may be arbitrary, any data that is not an inner product space gives a model of ¬REP-VSLIN, and m is not needed.
   - *No vector space is fixed, even up to isomorphism.* This is false: by Wigner's theorem the Born rule on $P(H)$ fixes H up to a unitary or anti-unitary map (not checked in Lean

for two different spaces; the Wigner theorem in `ForPhyslib` is for maps of one space to itself). For $m(H)$ in particular, $m(H) \cong H$ by construction (`twistedIsometry`).

- *The linear structure is not fixed by the rays and the Born rule.* This is proved (`exists_second_linear_structure`): the carrier of H has a second complex inner product space structure, carried over along m, with the same scalar multiplication, hence the same rays, and the same Born rule, but a different addition. It uses eqs. (3), (4) and (6), and needs $\dim H \geq 2$ (`map_add_of_map_smul`).

The step of the proof that breaks under the first two readings is "since the map is non-linear, $H \not\equiv m(H)$ in the category of vector spaces", which is false (Log, "Problems found"). The third reading is proposed as the precise statement. The item stays open until the brief adopts it, or another one, and corrects that sentence. (Log, "Differences from the brief".)